A Business Intelligence Approach to Retail Expansion

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Introduction

The United States consists of 1.9 billion acres of land. Of that staggering amount, millions of acres remain uninhabited and unused (National Public Radio, 2019). Given the struggles that much of the unemployed labor force has had in finding work (Center for American Progress, 2025), and the resulting decline in casual spending (McKinsey & Company, 2026), I was inspired to investigate what businesses can do to create jobs while profiting as well. During this exploration, I became interested in the tug-of-war happening between online and in-person shopping (Raj & Sharma, 2025). Despite the rising pervasiveness of e-commerce, brick-and-mortar stores continue to dominate in certain respects. According to business.com; instant gratification, customer service, and social engagement are some of the reasons that stores have been able to hold their own among customers (Dublino, 2026). On the business side, retailers also benefit from physical stores as products bought from them are roughly 16% less likely to be returned. And in an America where retail returns totaled $849.9 billion in 2025, that is a significant margin (Capital One Shopping, 2026). Given this information, I believe that more businesses should consider investing in physical stores and physical advertising. However, such a large multi-faceted investment should be made strategically in the regions that would yield the most profit. This led to the development of my research question:

For a national apparel retailer, which U.S. states represent the strongest opportunities for store expansion, inventory redistribution and marketing investment based on retail sales growth and economic stability?

The phrase "strongest opportunities" can be further distilled into the following research questions:

1. *Which US states are experiencing the most revenue growth?*
2. *Which states have the most potential, having low unemployment and high revenue growth?*
3. *Which states would be a bad investment, having high unemployment and low revenue growth?*

This project benefits stakeholders by providing a multi-dimensional framework for investment and resource allocation. Instead of viewing the United States as a monolith, this analysis utilizes data abstraction to categorize states into strategic zones, allowing for more nuanced decision-making that benefits both business and the labor force.

Datasets

To conduct this analysis, I utilized two primary, publicly available datasets from the U.S. government: the U.S. Census Bureau and the U.S. Bureau of Labor Statistics (BLS). These organizations are federally funded and trusted sources for objective quantitative data about the country's economy. These datasets were selected to represent economic stability (via unemployment rate) and revenue potential (via retail sales) because they provide a comprehensive view of state-level consumer activity and labor conditions.

Dataset 1: Monthly State Retail Sales (U.S. Census Bureau)

The primary dependent variable, Retail Growth %, was derived from the U.S. Census Bureau's Monthly State Retail Sales report, which is designed to track consumer spending patterns across the 50 states and the District of Columbia. However, I was surprised to find that substantial preprocessing was required to make this dataset analytically usable. The original data was structured in a wide format, with each month represented as a separate column instead of having Date as one column. I transformed this into a long format using the Pivot function in Tableau. Additionally, I filtered the dataset to include only the "TOTAL" NAICS category to avoid double counting across retail sub-industries. State abbreviations were converted to full state names to support geographic mapping and ensure consistency with the secondary dataset. Lastly, I derived a standardized monthly date field to preserve temporal consistency.

Dataset 2: Monthly State Unemployment Rate (Bureau of Labor Statistics)

The primary independent variable, Unemployment Rate, was sourced from the BLS, which produces labor market indicators to inform policy and economic analysis. The raw dataset contained multiple demographic and labor force metrics across 1,785 rows and 11 columns. To align with the scope of this project, I reduced the dataset to three key variables: State, Date, and Unemployment Rate. I derived another unified monthly date attribute from separate year and month columns to match the one I made in the retail dataset. Additionally, I removed non-state

entries (e.g., city-level observations) to ensure consistency across all 51 geographic cases. One limitation is that data for October 2025 was unavailable, so this period was excluded from both datasets to maintain their alignment.

Data Integration and Variables

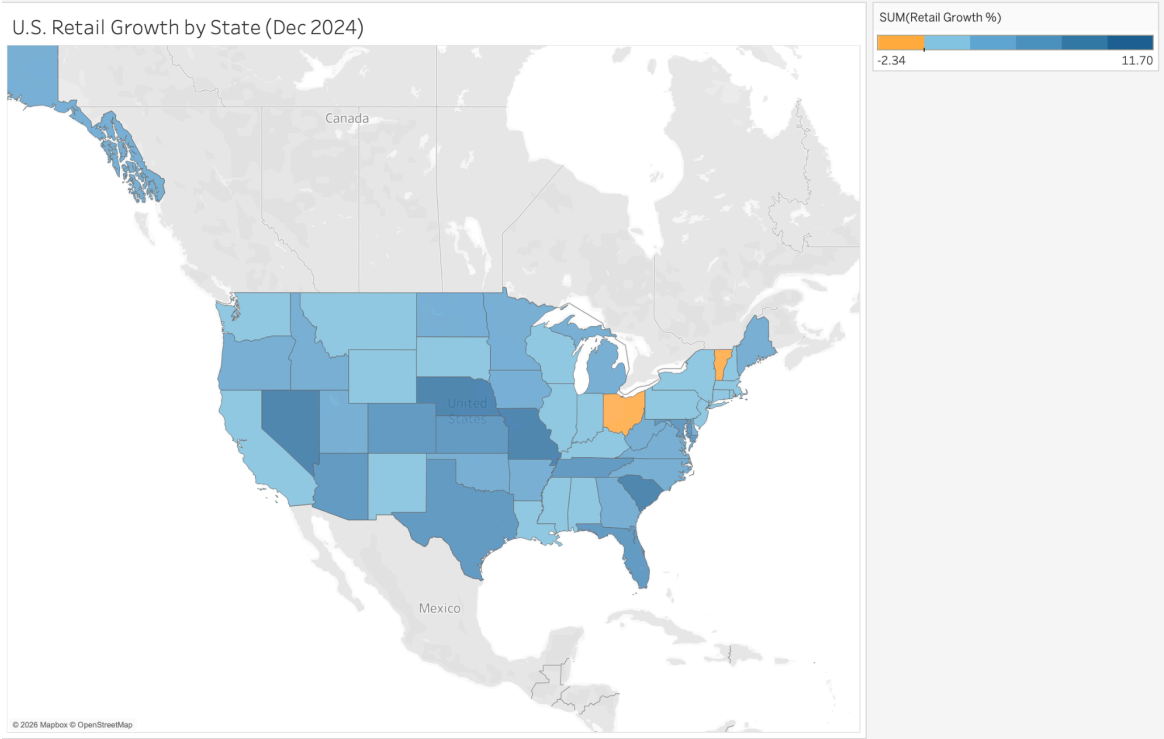
The datasets were integrated using a Relationship in Tableau using the State and Date shared keys. Key variables in the final database include:

- Retail Growth % (Quantitative/Derived): Monthly percentage change in retail sales, representing consumer demand and economic performance.
- Unemployment Rate (Quantitative): A measure of economic stability or risk.
- State (Categorical/Spatial)
- Date (Temporal)
- Category (Categorical/Derived): A derived attribute I made to group states based on their investment potential.
    - Low Unemployment + High Retail Growth - High Potential
    - Low Unemployment + Low Retail Growth - Worth Exploring
    - High Unemployment + Low Retail Growth - Too Risky

These preprocessing and integration steps ensured that both datasets were aligned in structure and granularity, allowing for accurate comparisons across states and over time. While minor limitations existed, such as missing observations and poor formatting, the final dataset provides a robust foundation for my analysis and eventual business recommendations.

Data Exploration

Chart 1: Choropleth Map of U.S. Retail Growth (Dec 2024)



Variables: Retail Growth % (Census, monthly % change in retail sales), State (geographic)

This map serves as the starting point of the analysis by answering a simple but critical question: *where is retail currently performing well?* Retail Growth % is derived from the Census Bureau and reflects month-over-month percent change in total retail sales (NAICS = TOTAL), allowing for consistent comparisons across states. I selected December 2024 because, according to the National Retail Federation, retail sales usually perform the best during the winter holiday season (Matthews, 2025). I thought this would be a good time to get a concise look at the population's attitudes towards spending on retail items which are often non-essential.

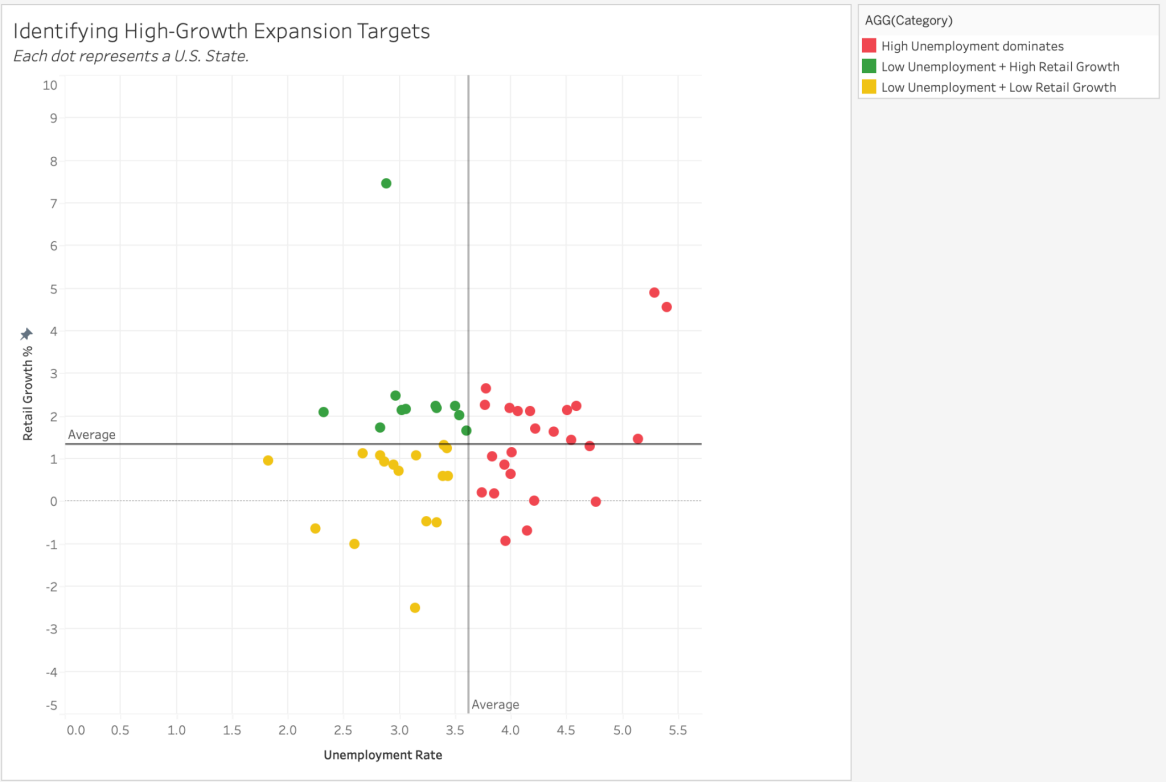
By encoding growth using a color gradient, the map highlights geographic patterns in consumer demand. Stronger growth appears clustered in parts of the West and Southeast, while weaker or negative growth is visible in select Midwestern states. This immediately reinforces the idea that the U.S. should not be treated as a single market.

Limitations:

Because this is a single-month snapshot (December 2024), it may reflect seasonal holiday effects rather than sustained performance.

This chart provides the primary spatial **overview.** By using a color gradient **(luminance)** to represent growth magnitude, it allows stakeholders to instantly identify geographic clusters of prosperity. For example, the dark blue hues in the West suggest a regional trend of resilience compared to the more volatile orange shades seen in the Midwest. I also used **hue** to encode the unemployment rates visible in the Toolkit upon hovering over each state. In doing so using red and green, I was able to intuitively communicate where each state's unemployment rate fell in comparison to the national average (which was another derived attribute I calculated from the data).

Chart 2: Relational Scatter Plot (Expansion Targets)



Variables:

Unemployment Rate (BLS, % of labor force), Retail Growth % (Census, % change), Category (derived grouping)

This chart answers the question: *which states balance strong demand with economic stability?* Each point represents a state, plotted using average unemployment rate and average retail growth over time.

The reference lines divide the chart into four strategic zones. States in the top-left quadrant (low unemployment, high growth) represent the strongest expansion targets, as they combine stable labor markets with strong consumer spending. Stated simply, there are people in these states with stable income and the will to spend it. In contrast, states in the bottom-right quadrant indicate high-risk environments where weak retail demand coincides with economic instability. Some of these high-risk states include **Louisiana, Rhode Island, and Ohio.** Ohio's presence in this list is corroborated by its stand-out appearance in the chloropleth map as a state with abnormally low retail growth, even during the holiday season.

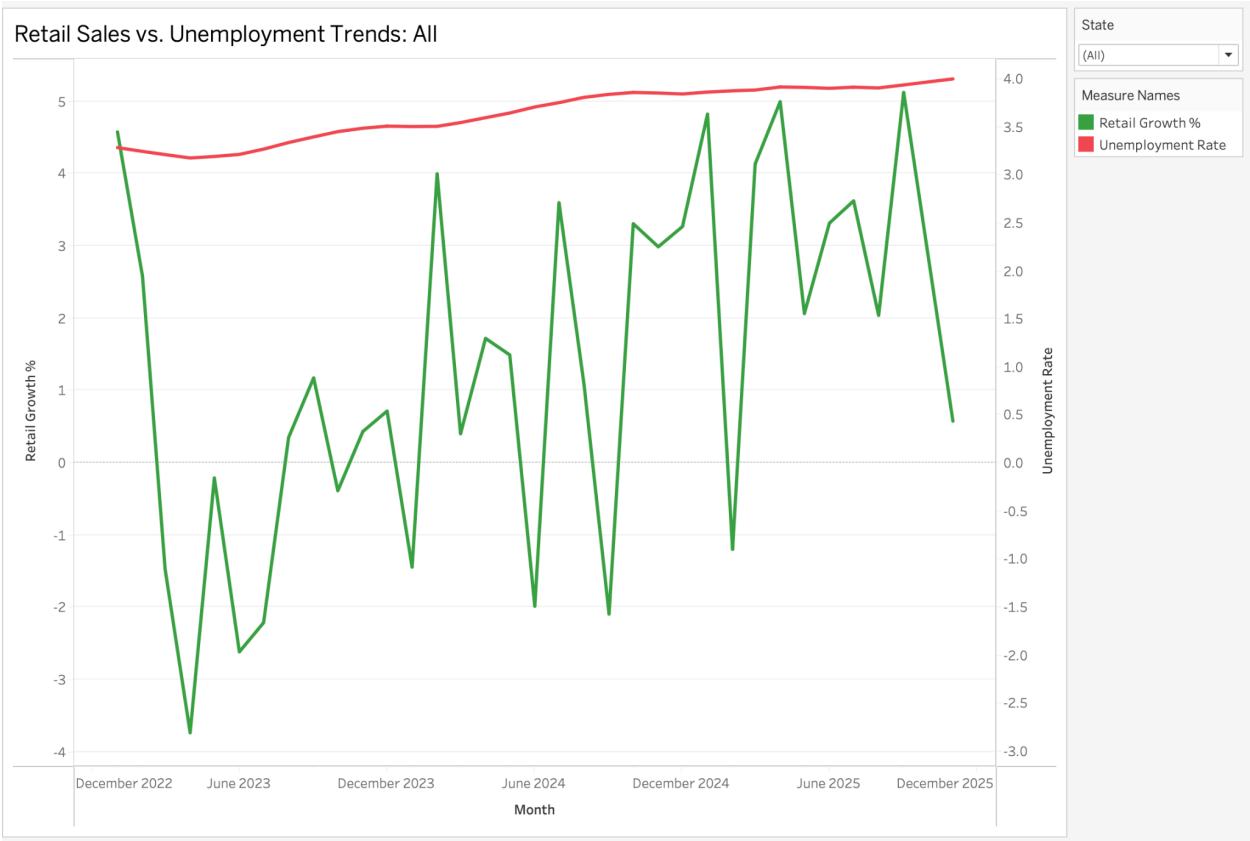
While there is a general trend that higher unemployment aligns with lower growth, the spread of points shows that this relationship is not absolute. Some states outperform expectations, suggesting additional local factors influence retail success.

This idiom addresses the task of **correlation and partitioning**. By plotting states based on two quantitative attributes, I have categorized them into strategic categories. The green dots represent the "Ideal" zone (Low Unemployment paired with High Growth) while red dots signal regions where high unemployment dominates, helping stakeholders prioritize investment targets. This use of hue is in line with Tamara Munzner's principles of effectiveness, utilizing the most accurate visual channels for the intended task. Overall, this is the most directly actionable chart, identifying which states are worth investing in versus avoiding.

Limitations:

Averaging across time may hide short-term opportunities or risks.

Chart 3: Dual-Axis Line Chart (Temporal Trends)



Variables:

Date (monthly), Retail Growth % (Census), Unemployment Rate (BLS)

While the previous charts focus on *where*, this chart answers *when*. It tracks how retail performance and unemployment evolve over time across all states. I also included a filter on the right side to view the chart for each of the states.

Retail growth shows significant volatility, with sharp spikes and drops, while unemployment follows a much smoother and gradual trend. This contrast suggests that retail performance reacts quickly to short-term conditions whereas unemployment reflects slower structural changes in the economy. This means that even in areas that appear economically stable, there can still be huge fluctuations in consumer spending and retail demand overall.

Limitations:

Dual-axis charts require careful interpretation, as the two variables operate on different scales. Additionally, this suggests that unemployment rate or even overall economic health are not enough to account for retail growth and therefore for revenue. However, I believe this is a reminder that this analysis is not intended as a final investment strategy, but rather as a resource to business professionals. Such stakeholders would likely be able to conduct a more exhaustive analysis that would inform any final company decisions. This chart highlights the importance of timing, showing that expansion decisions should account for volatility, not just averages.

Chart 4: Retail & Unemployment Lookup Table

Retail Growth & Unemployment Rates by US State (2023-2025)

State	Retail Growth %	Unemployment Rate
Alabama	0.917	2.865
Alaska	2.129	4.503
Arizona	2.254	3.768
Arkansas	0.591	3.435
California	1.469	5.138
Colorado	2.197	3.994
Connecticut	2.194	3.338
Delaware	0.189	3.853
District of Columbia	4.903	5.285
Florida	2.226	3.326
Georgia	1.237	3.429
Hawaii	7.466	2.882
Idaho	2.243	3.506
Illinois	1.280	4.712
Indiana	1.040	3.829
Iowa	1.071	3.150
Kansas	0.586	3.388
Kentucky	-0.011	4.768
Louisiana	-0.700	4.150
Maine	0.714	2.994
Maryland	1.731	2.832
Massachusetts	0.634	4.003
Michigan	2.234	4.591
Minnesota	2.163	3.056
Mississippi	-0.509	3.338
Missouri	2.017	3.538
Montana	1.071	2.832
Nebraska	1.131	2.671
Nevada	4.560	5.397
New Hampshire	-0.997	2.594
New Jersey	1.440	4.547
New Mexico	1.149	4.006
New York	2.111	4.176
North Carolina	1.643	3.600

AVG(Unemployment R...)
1.826 5.397

AVG(Retail Growth %)
-2.497 7.466

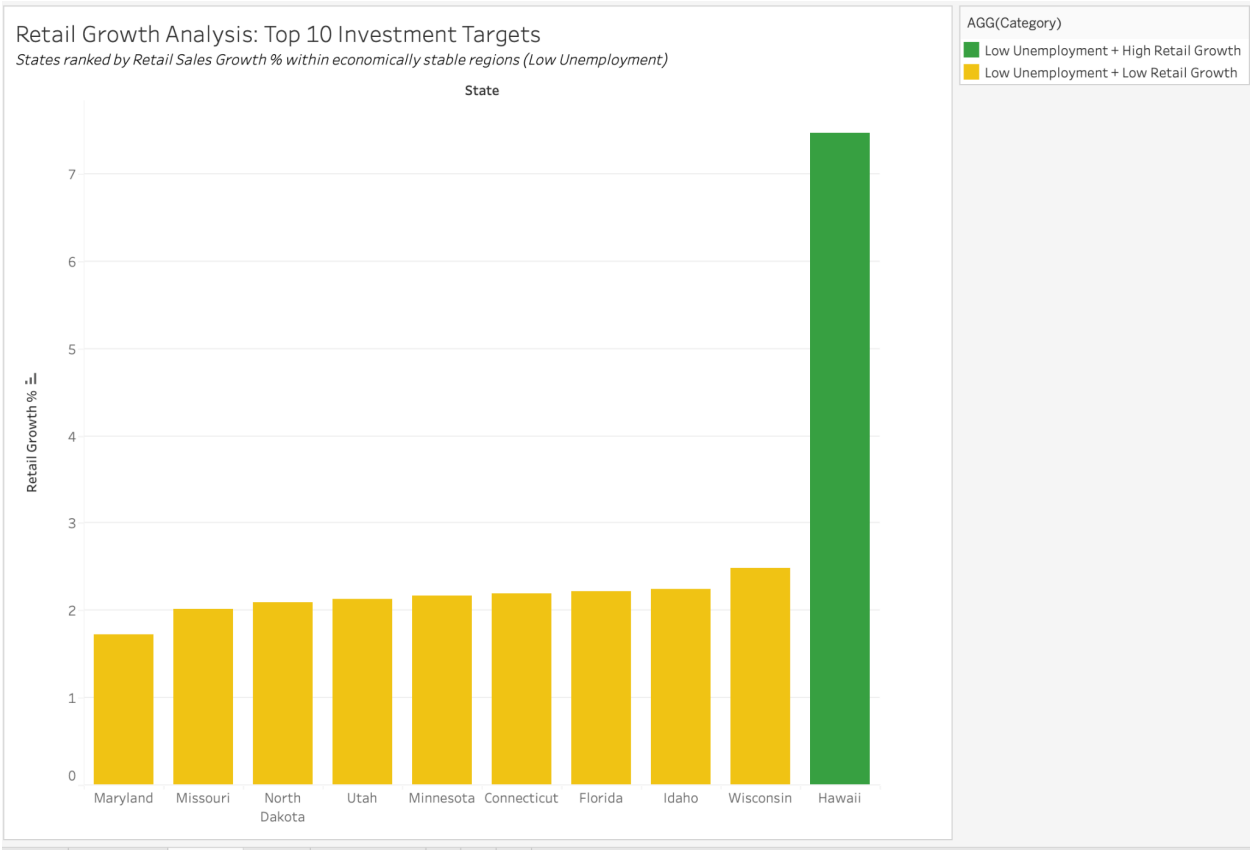
Variables:

State, Retail Growth % (Census), Unemployment Rate (BLS)

This table provides detailed, state-level values to support and validate insights from other charts. Retail Growth % and Unemployment Rate are displayed side by side, with **color encoding used to emphasize magnitude** (green for stronger growth, red for higher unemployment).

This view makes it easy to identify standout states. For example, **Hawaii** shows exceptionally high retail growth (7.46%) paired with moderate unemployment, reinforcing its classification as a high-potential market. I created this table to allow for quick lookup of data, supporting decision-making by allowing stakeholders to verify specific information they are seeking before making investment choices.

Chart 5: Top 10 Investment Targets (Bar Chart)



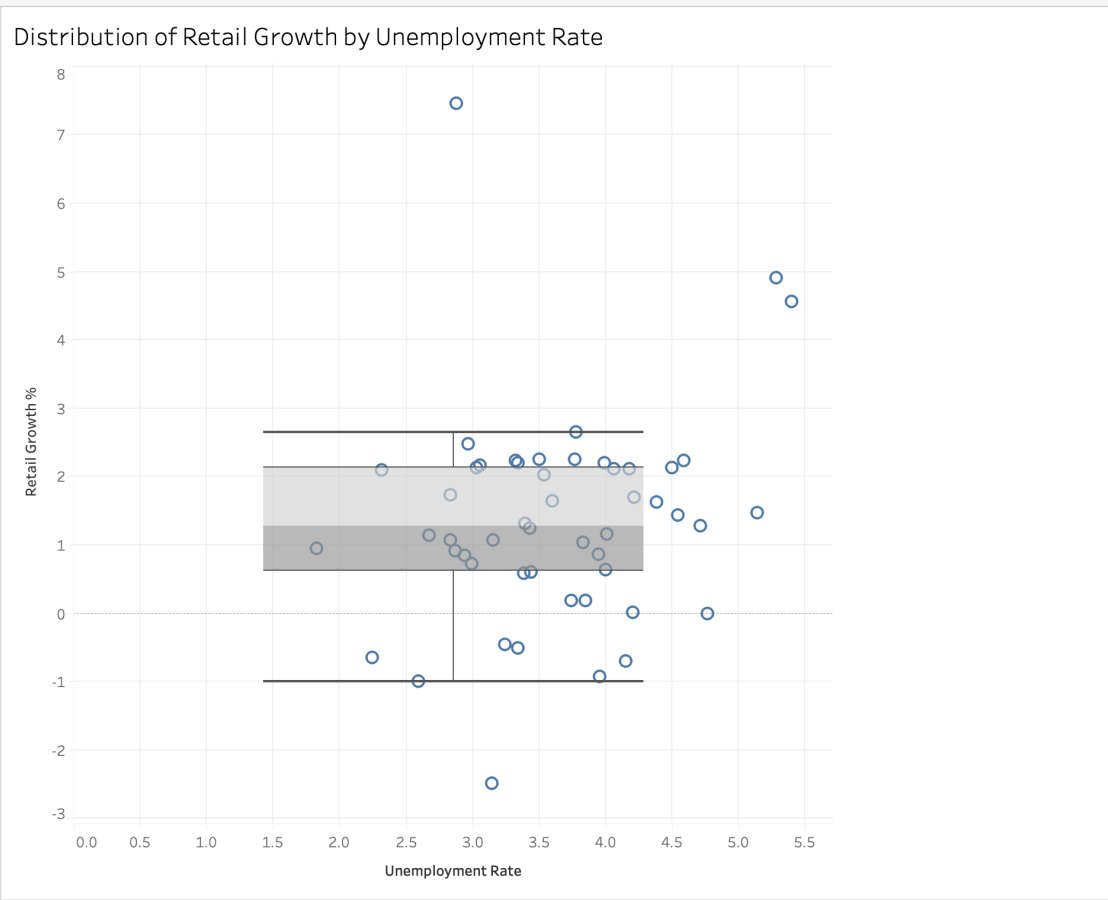
Variables:

State (categorical), Retail Growth % (Census)

This chart translates analysis into clear action by ranking the top 10 states based on retail growth within economically stable regions (low unemployment, encoded by color). Using **aligned length, the most accurate visual encoding for comparison**, it clearly shows relative differences in performance.

Hawaii stands out as a clear outlier, significantly outperforming all other states. The remaining top states show more moderate but consistent growth, indicating a tier of strong but less extreme opportunities that are nevertheless worth investigating. This chart directly answers where to allocate resources first, prioritizing the highest-return opportunities.

Chart 6: Distribution of Retail Growth by Unemployment Rate



Variables:

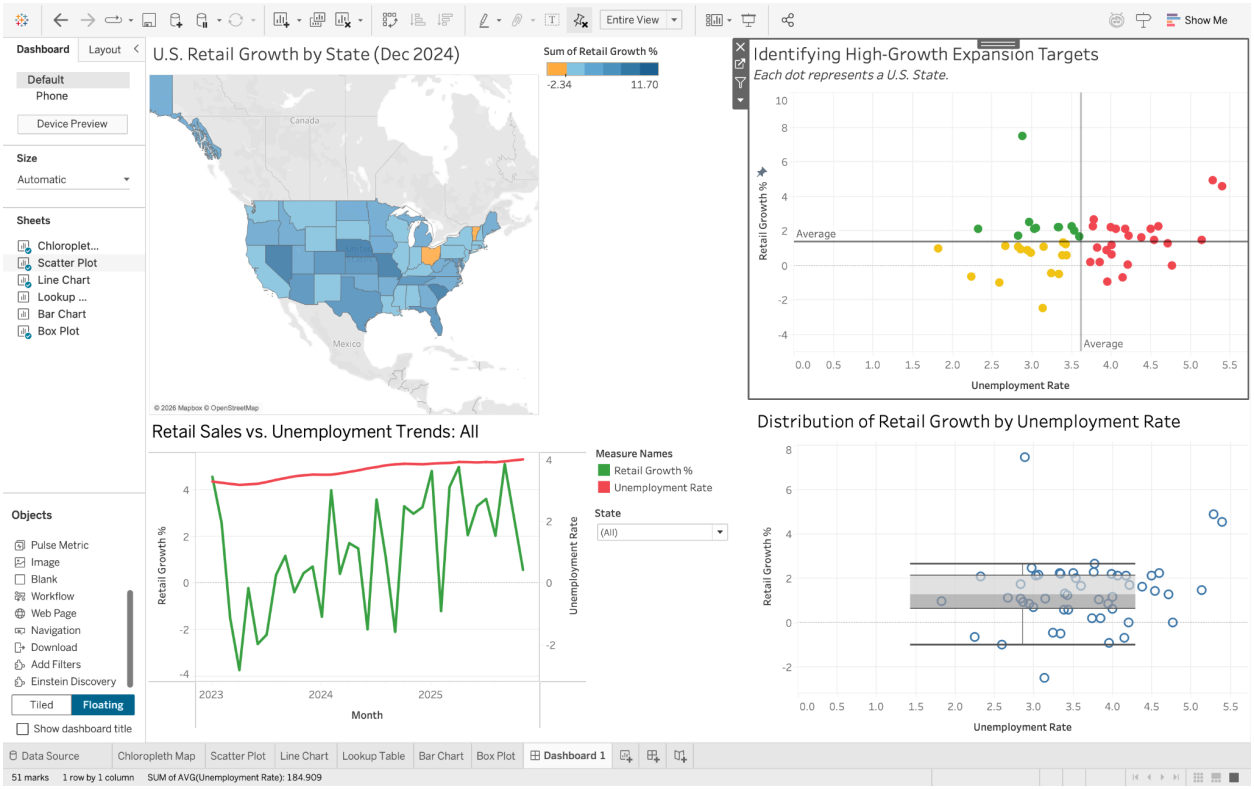
Retail Growth % (Census), Unemployment Rate (BLS)

This chart adds depth by examining the overall distribution of retail growth across different unemployment levels. While the scatter plot shows individual states, this view summarizes variability using median, quartiles, and outliers. The wide spread of values and presence of outliers indicate that retail performance varies significantly even among states with similar unemployment rates. This reiterates that unemployment alone is not sufficient to predict retail success. The box plot also revealed more states that may be worth exploring, some of which did not stand out in the other visualizations. These include **Washington DC, Nevada, and Wisconsin.** This chart reinforces the need for a multi-factor approach, showing that successful expansion decisions require more than a single economic indicator.

Limitations:

Grouping and summarization reduce granularity and may mask individual state behavior.

Summary & Conclusions



The final dashboard brings these views together into a single decision-making tool. The map provides a geographic entry point, while the scatter plot and bar chart identify high-priority

markets. The line chart and box plot add context about behavior over time and relative distribution, showing how performance evolves and varies across states.

Based on this analysis, stakeholders are encouraged to explore expansion in states like **Hawaii, Wisconsin, Idaho, Florida and Washington DC** while approaching states like **Ohio, Rhode Island, and Louisiana** with more caution. Together, these visualizations tell a clear story: the United States is not a uniform market, and retail success depends on both location and economic conditions. While unemployment provides a useful baseline for assessing risk, it does not fully determine retail performance. Several states demonstrate strong growth despite moderate unemployment, highlighting opportunities that might otherwise be overlooked.

For a national retailer, this means that expansion should be targeted not just toward economically stable regions, but toward states that combine stability with demonstrated consumer demand. This approach supports more efficient allocation of stores, inventory, and marketing resources while also improving economic opportunities for citizens across the country.

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